Natural Language Processing

Assignment 2

Conversational AI System

Nawaal

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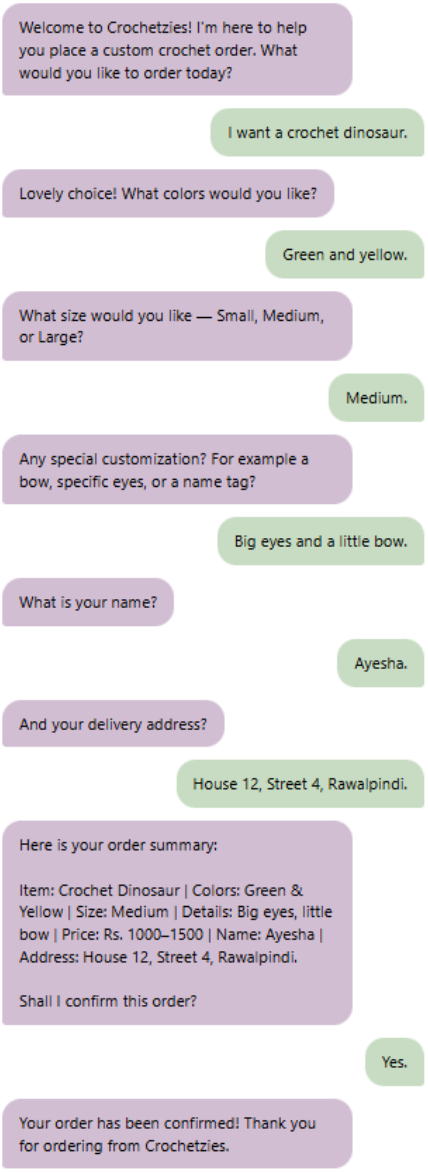
Use Case Description

Crochetzies is a small handmade crochet business that specializes in multiple crochet items like characters, animals, and a wide range of other custom designs. The Crochetzies Chatbot serves as a virtual order-taking assistant that guides customers through placing a custom crochet order. The Chatbot collects necessary details, including the item type, item customization preferences such as the color, the design, and the size of the item. Based on the data that it collects, it estimates a value in PKR. Finally, it collects the customer's name and address and then confirms the order with the order details before closing the conversation.

Example Dialogues

Dialogue 1 - Happy Path



Dialogue 2 - Uncertain Customer

Welcome to Crochetzies! What would you like to order today?

I'm not sure, what can you make?

We can make characters, animals, plushies, keychains, and more! Do any of those interest you?

Maybe a cat?

A crochet cat sounds adorable! What colors would you like?

Pink and white.

What size would you like — Small, Medium, or Large?

Small please.

Any special details you'd like to add?

No that's fine.

What is your name?

Sara.

And your delivery address?

Flat 5, Block B, Lahore.

Here is your order summary:

Item: Crochet Cat | Colors: Pink & White | Size: Small | Price: Rs. 500–800 | Name: Sara | Address: Flat 5, Block B, Lahore.

Shall I confirm this order?

Yes please!

Your order has been confirmed! Thank you for ordering from Crochetzies.

Dialogue 3 - Customer Changes Their Mind

Welcome to Crochetzies! What would you like to order today?

I want a crochet bear, large, brown and black.

Great choice! Any special customization details?

Actually can I change the size to medium?

Of course! Medium it is. Any customization details?

No just keep it simple.

What is your name?

Ali.

And your delivery address?

House 3, Model Town, Lahore.

Here is your order summary:

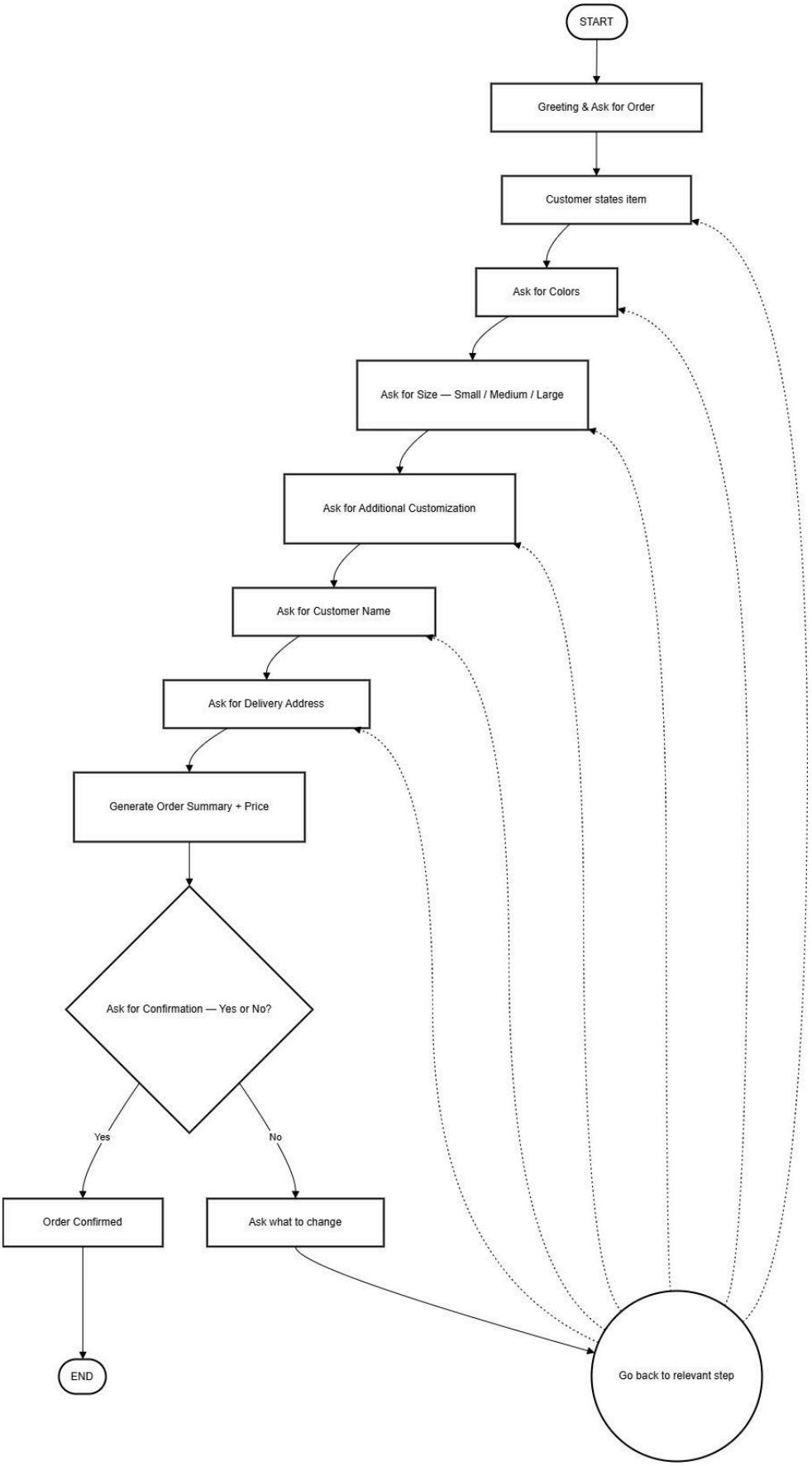
Item: Crochet Bear | Colors: Brown & Black | Size: Medium | Price: Rs. 1000–1500 | Name: Ali | Address: House 3, Model Town, Lahore.

Shall I confirm this order?

Yes.

Your order has been confirmed! Thank you for ordering from Crochetzies.

Conversation flow diagram



Context memory management scheme

Strategy: Hybrid (Sliding Window + Summarization)

The chatbot uses a hybrid memory strategy. The last 10 turns of conversation are always kept in full. If the conversation exceeds 10 turns, older turns are compressed into a short summary before being dropped. This ensures no critical order details are ever lost while keeping the context window manageable.

Parameter	Value
Max turns kept in full	10 turns
Summary trigger	Conversation exceeds 10 turns
Summary max length	3–4 sentences
Context sent to model	Summary (if any) + last 10 turns + current message

Context Structure (what is sent to the model each turn):

[SYSTEM PROMPT]
You are a friendly order-taking assistant for Crochetzies, a custom crochet business. Guide the customer through placing a custom order. Collect items, colors, size, customization, name, and delivery address. Calculate price based on size in PKR. Do not go off topic.

[SUMMARY] (only if conversation exceeds 10 turns)
e.g. "The customer wants a medium green and yellow crochet dinosaur with big eyes and a bow."

[CONVERSATION HISTORY]
Last 10 turns of dialogue

[CURRENT USER MESSAGE]
User: ...

Signal vs Noise:

Signal (kept): item type, colors, size, customization details, customer name, delivery address, any changes made mid-conversation.
Noise (dropped/compressed): greetings, filler responses like "ok" or "yes please", repeated bot questions, off-topic messages.

Inference latency benchmarks

Model: Qwen 2.5 1.5B (Q4_K_M, via Ollama)
Inference: Ollama
Hardware: CPU only, 16GB RAM
Tool: PowerShell Measure-Command

Prompt Type	Description	Response Time
Short	"Hi"	1.47 seconds
Medium	Custom order request	7.44 seconds
Long	Full order with all details	9.49 seconds

Memory Usage: 8.9GB RAM while model is running.